Vakeel Saab App — SRS & Technical Blueprint

Production-Grade Specification for AI Coding Agents & Development Teams

Flutter (Dart) | Supabase (PostgreSQL 18) | Dual-Tier Verification | Zero-Cost Serverless

1. Executive Summary & Product Architecture

The **Vakeel Saab App** is a secure, private, and democratic community mobile and web application engineered exclusively for legal professionals enrolled under the Advocates Act, 1961. Built on a serverless Backend-as-a-Service (BaaS) architecture using **Supabase** and **Flutter (Dart)**, the platform optimizes infrastructure costs while ensuring rigid professional trust and compliance with Bar Council of India (BCI) ethical guidelines.

Core Infrastructure Stack:

- **Frontend Mobile & Web:** Single Flutter codebase responsive for Android, iOS, and Web Admin dashboard with `go_router` deep-linking.
- **Backend & Database:** Supabase PostgreSQL 18 with custom enums, trigram full-text search, and Row-Level Security (RLS) enforcement.
- **Authentication & SMS:** Firebase Auth Phone Sign-in or MSG91 OTP with secure session management.
- **Push Notifications:** Firebase Cloud Messaging (FCM) dispatched via Supabase Edge Functions / DB Webhooks.

2. Complete Core Feature Matrix

Module	Core Features & Functionality	Access Rules & Permissions
Authentication & Onboarding	Mobile OTP authentication, Joining Request form (Name, DOB, Gender, State Bar Association, Enrollment #, Place of Practice, Courts of Practice), and mandatory BCI Terms of Service agreement checkbox.	Open to all registering legal advocates.

Module	Core Features & Functionality	Access Rules & Permissions
Verification & Trust (Dual-Tier)	**Tier 1 (Approved Member):** Admin manual approval of basic registration. **Tier 2 (Verified Advocate):** Mandatory/Optional Bar ID Card image upload; human admin verification matches ID with enrollment number to award verified gold/blue badge.	**Tier 1:** Wall, Directory, Contacts. **Tier 2:** Elections, poll creation, and premium query rooms.
Community Wall	Paginated feed (`created_at DESC`), create/update/delete text and media posts, optimistic likes, comment threads, author profile inspection.	Approved Members (Tier 1 & Tier 2).
Report & Moderation	3-dot overflow menu on posts/comments to flag unprofessional or inappropriate content. Admin review hub for content take-downs and user warnings.	Approved Members (reporting), Admins (moderation).
Broadcast Channel	One-way admin announcement channel with priority flags (`normal`, `urgent`, `strike_notice`) coupled with FCM push alerts.	**Creation:** Admins only. **Viewing:** All approved members.
Directory & Search	Global debounced search (`pg_trgm` full-text search) by name, enrollment number, or place of practice. Filter by Courts of Practice.	Approved Members.
Privacy Phone Requests	Phone numbers masked by default (`+91 XXXXX XXXXX`). Advocates submit access requests; target user accepts/rejects to unmask.	Approved Members.
Elections & Polls	Scheduled elections, candidate nominations with manifestos, wall voting polls, anonymous duplicate-checked voting, and auto-reveal results.	**Nominations & Voting:** Tier 2 Verified Advocates only.
Directory of Contacts	Reference desk for Bar Association office bearers, court registry desks, legal aid, and emergency helplines.	Approved Members.

3. Production PostgreSQL Schema & RLS Implementation

Execute the following script inside the Supabase SQL Editor to initialize the database architecture, constraints, and Row-Level Security policies:

```sql
-- 1. ENUMS & EXTENSIONS
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";

CREATE TYPE user_status AS ENUM ('pending', 'approved', 'rejected');
CREATE TYPE request_status AS ENUM ('pending', 'accepted', 'rejected');
CREATE TYPE report_target_type AS ENUM ('post', 'comment', 'user');
CREATE TYPE report_status AS ENUM ('pending', 'reviewed', 'dismissed', 'action_taken');
CREATE TYPE election_status AS ENUM ('upcoming', 'nominations_open', 'voting_open', 'closed');

-- 2. USERS TABLE
CREATE TABLE public.users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    auth_user_id UUID UNIQUE REFERENCES auth.users(id) ON DELETE CASCADE,
    phone VARCHAR(20) NOT NULL UNIQUE,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    dob DATE NOT NULL,
    gender VARCHAR(20) NOT NULL,
    state_bar_association VARCHAR(150) NOT NULL,
    enrollment_number VARCHAR(100) NOT NULL UNIQUE,
    place_of_practice VARCHAR(150) NOT NULL,
    courts_of_practice TEXT[] DEFAULT '{}',
    bio TEXT,
    avatar_url TEXT,
    status user_status DEFAULT 'pending',
    is_verified BOOLEAN DEFAULT FALSE,
    bar_id_card_url TEXT,
    is_admin BOOLEAN DEFAULT FALSE,
    phone_visibility BOOLEAN DEFAULT FALSE,
    terms_accepted BOOLEAN DEFAULT FALSE,
    terms_accepted_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- 3. VOTES TABLE (WITH ANTI-DOUBLE VOTING CONSTRAINT)
CREATE TABLE public.votes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    election_id UUID NOT NULL REFERENCES public.elections(id) ON DELETE CASCADE,
    position_name VARCHAR(100) NOT NULL,
    voter_id UUID NOT NULL REFERENCES public.users(id) ON DELETE CASCADE,
    candidate_id UUID NOT NULL REFERENCES public.users(id) ON DELETE CASCADE,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    CONSTRAINT unique_vote_per_position UNIQUE (election_id, position_name, voter_id)
);
```

4. Frontend Flutter Architecture & UI Design System

Design System & Theming

- **Primary Color (Judicial Navy):** `#0A192F` (App bars, primary buttons, branding).

- **Secondary Color (Brass Gold):** `#D4AF37` (Tier 2 verified badges, elections, accents).
- **Typography:** Google Fonts *Plus Jakarta Sans* for structured corporate readability.
- **Feedback Alerts:** Emerald Green (`#10B981`) for success, Amber (`#F59E0B`) for pending states, and Rose (`#EF4444`) for danger/reports.

Admin Web Dashboard Layout

The web portal uses a responsive master-detail split layout:

- **Left Navigation Sidebar:** Quick switching between Pending Join Requests, Content Moderation Queue, Broadcast Center, and Elections.
- **Right Inspection Pane:** Side-by-side verification review showing advocate metadata and an expandable high-res preview of the uploaded Bar ID card image.

5. Phased Implementation Roadmap

- **Phase 1: Foundation (Weeks 1–2) —** Supabase schema initialization, RLS policies, Flutter project structure, GoRouter routing, and Phone OTP auth.
- **Phase 2: Onboarding & Admin Review (Weeks 3–4) —** Joining Request form, Bar ID Card upload, legal checkboxes, and Admin Web Verification queue.
- **Phase 3: Community Wall & Directory (Weeks 5–6) —** Paginated wall feed, likes/comments, global directory search, and phone access request workflow.
- **Phase 4: Elections, Broadcasts & Polish (Weeks 7–8) —** Election lifecycle, candidate nominations, secure voting constraints, FCM push alerts, and beta testing.